

Arjuna JEE (2025)

Mathematics

Binomial Theorem

DPP: 2

- Q1** The coefficient of the middle term in the expansion $(x + 2y)^6$ is
 (A) 6C_3
 (B) $8({}^6C_3)$
 (C) $8({}^6C_4)$
 (D) 6C_4
- Q2** If in the expansion of $(1 + x)^m(1 - x)^n$, the coefficient of x and x^2 are 3 and -6 respectively, then m is
 (A) 6 (B) 9
 (C) 12 (D) 24
- Q3** If in the expansion of $\left(\sqrt[3]{2} + \frac{1}{\sqrt[3]{3}}\right)^n$, the ratio of the 7^{th} term from the beginning to the 7^{th} term from end is equal to $1/6$ then n is equal to
 (A) 3
 (B) 6
 (C) 9
 (D) None of these
- Q4** The coefficient of the term independent of x in the expansion of $\left(\frac{x+1}{x^{2/3}-x^{1/3}+1} - \frac{x-1}{x-x^{1/2}}\right)^{10}$ is
 (A) 210 (B) 105
 (C) 70 (D) 112
- Q5** The coefficient of x^7 in the expansion of $(1 - x - x^2 + x^3)^6$ is
 (A) -132 (B) -144
 (C) 132 (D) 144
- Q6** If the greatest value of the term independent of ' x ' in the expansion of $(x \sin \alpha + a \frac{\cos \alpha}{x})^{10}$ is $\frac{10!}{(5!)^2}$, then the value of ' a ' is equal to
 (A) -1 (B) 1
 (C) -2 (D) 2
- Q7** The coefficients of three consecutive terms of $(1 + x)^{n+5}$ are in the ratio 5 : 10 : 14. Then $n =$ _____.
- Q8** The total number of irrational terms in the binomial expansion of $(7^{1/5} - 3^{1/10})^{60}$ is
 (A) 48 (B) 49
 (C) 54 (D) 55
- Q9** The coefficient of the term independent of x in the expansion of $(1 + x + 2x^3)\left(\frac{3}{2}x^2 - \frac{1}{3x}\right)^9$, is
 (A) $1/3$ (B) $19/54$
 (C) $17/54$ (D) $1/4$
- Q10** The term independent of x in $(1 + x)^m(1 + \frac{1}{x})^n$
 (A) ${}^{m+n}C_n$
 (B) ${}^{m+n}C_{n+1}$
 (C) ${}^{m+n}C_{m-n}$
 (D) None of these
- Q11** The middle term in the expansion of $\left(x^2 + \frac{1}{x^2} + 2\right)^n$ is
 (A) $\frac{n!}{\left(\frac{n}{2}!\right)^2}$
 (B) $\frac{(2n)!}{\left(\left(\frac{n}{2}\right)!\right)^2}$
 (C) $\frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{n!} \cdot 2^n$
 (D) $\frac{1 \cdot 3 \cdot 5 \cdots (2n+1)}{n!} \cdot 2^n$
- Q12** The number of terms which are free from radical signs in the expansion of $(y^{1/5} + x^{1/10})^{55}$ is
 (A) 5 (B) 6
 (C) 7 (D) None
- Q13**



The number of integral terms in the expansion of $(\sqrt{3} + \sqrt[8]{5})^{256}$ is

- (A) 32 (B) 33
(C) 34 (D) 35

Q14 The sum of the rational terms in the binomial

expansion of $(2^{1/2} + 3^{1/5})^{10}$ is

- (A) 25 (B) 32
(C) 9 (D) 41

Q15 In the expansion of $(5^{1/2} + 7^{1/8})^{1024}$ the number of integral term is

- (A) 128 (B) 129
(C) 130 (D) 131

Q16 A ratio of 5th term from the beginning to the 5th term from the end in the binomial expansion

of $(2^{1/3} + \frac{1}{2(3)^{1/3}})^{10}$ is

- (A) $1 : 2(6)^{1/3}$
(B) $1 : 4(16)^{1/3}$
(C) $4(36)^{1/3} : 1$
(D) $2(36)^{1/3} : 1$

Q17 If the 9th term in the expansion of $(\frac{1}{x^2} + \frac{x}{2} \log_2 x)^{12}$ is equal to 495, then the value of x is

- (A) 1 (B) 2
(C) 3 (D) 4

Q18 In the expansion of $(2^{1/5} + 7^{1/7})^{105}$, which of the following holds goods?

- (A) Number of rational terms are 4
(B) Number of irrational terms are 102
(C) Exactly one middle term is irrational
(D) Both middle terms are irrational

Q19 Let P be the 7th term from the beginning and Q be the 7th term from the end in the expansion of $(\sqrt[3]{3} + \frac{1}{\sqrt[3]{4}})^n$ where $n \in N$. If $12P = Q$, then find the value of n .

Q20 If the number of integral terms in the expansion of $(3^{1/2} + 5^{1/8})^n$ is exactly 33, then the least value of n is

- (A) 264 (B) 128
(C) 256 (D) 248



Answer Key

Q1 (B)
Q2 (C)
Q3 (C)
Q4 (A)
Q5 (B)
Q6 (D)
Q7 6
Q8 (C)
Q9 (C)
Q10 (A)

Q11 (C)
Q12 (B)
Q13 (B)
Q14 (D)
Q15 (B)
Q16 (C)
Q17 (D)
Q18 (A, B, D)
Q19 9
Q20 (C)



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